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Helena Fernandes Evaristo

Machine Learning Guided Adaptive Scheduling for Multi-core Real-time Operating Systems

Florianópolis
2027

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Dissertação submetida ao Programa de Pós-Graduação em Ciência da Computação da Universidade Federal de Santa Catarina para a obtenção do título de Mestre em Ciência da Computação.
Supervisor:: Prof. Antônio Augusto Medeiros Fröhlich, Dr.

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Prof.(a) xxxx, Dr(a).
Instituição xxxx

Prof.(a) xxxx, Dr(a).
Instituição xxxx

Prof.(a) xxxx, Dr(a).
Instituição xxxx

Certificamos que esta é a **versão original e final** do trabalho de conclusão que foi julgado adequado para obtenção do título de Mestre em Ciência da Computação.

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Coordenação do Programa de
Pós-Graduação

_____

Prof. Antônio Augusto Medeiros Fröhlich,
Dr.
Supervisor:

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ABSTRACT

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ABSTRACT

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ML	Machine Learning
OS	Operating System
PMC	Performance Monitoring Counter
RTOS	Real-time Operating Systems
WCET	Worst-case Execution Time

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1 INTRODUCTION

Currently, several areas of technology require software that must execute in real time. This often includes autonomous vehicles, aviation, robotics, industrial systems, and other critical systems where the software is directly responsible for system safety and control. In these domains, a failure to meet timing constraints during execution can lead to severe consequences. Thus, to ensure predictable execution and meet these timing requirements, Real-time Operating Systems (RTOS) are utilized. These systems are commonly employed in embedded scenarios, where other variables must also be considered, such as energy consumption (Rahim; Nair; Devaraj, 2025). When timing and energy consumption constraints are combined with the current widespread adoption of multicore systems, the algorithms responsible for scheduling and load balancing face a considerable challenge (Islam *et al.*, 2018; Hoffmann; Fröhlich, 2021). This scenario motivates the need for predictive algorithms that can capture the state of the system and estimate relevant characteristics of each task in order to schedule and balance them accordingly.

One of the major challenges introduced by multicore processors is interference, which arises from task contention for shared resources such as main memory, caches, and I/O peripherals. This contention introduces non-deterministic delays into task execution, making it difficult to accurately estimate the Worst-case Execution Time (WCET). Such interference affects the scheduler's ability to guarantee that tasks will meet their deadlines, thus compromising system reliability (Gracioli *et al.*, 2019; Horstmann; Hoffmann; Fröhlich, 2019; Aceituno *et al.*, 2023; Lugo *et al.*, 2022). Heterogeneous multicore processors introduce additional challenges as well, since they present cores with different characteristics, demanding more sophisticated schedulers and closer attention from load balancing mechanisms. The dynamism and variation of tasks' workloads are another challenge, as tasks may evolve over time, changing their resource utilization and demands and requiring adaptive schedulers (Mrabet; Karamti; Mahfoudhi, 2024; Jadon *et al.*, 2024).

In recent years, several works have proposed the use of Performance Monitoring Counters (PMCs) in predictive algorithms, relying on hardware information to guide decisions. Most of these approaches rely on Machine Learning (ML) techniques, however, they also report difficulties and pitfalls that hinder their adoption. ML solutions usually introduce prohibitive overhead and fail to generalize well to new task-sets and platforms (Rahim; Nair; Devaraj, 2025). Moreover, the use of PMCs makes it difficult to design architecture-agnostic predictive algorithms. Although some works propose solutions to this issue, they remain limited by problems such as the divergence of PMC semantics across architectures, lack of counters in some platform implementations and also noise introduced by the Operating System (OS) and other external sources (Das *et al.*, 2019).

Considering this scenario, this dissertation addresses the main problems associated with the use of PMCs in ML algorithms, offering an analysis based on data collected from a RTOS running on RISC-V and ARM platforms. We also propose an architecture-agnostic solution that overcomes the identified problems and evaluate its performance using several real-time task-sets.

1.1 OBJECTIVES

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1.1.1 General Objectives

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1.1.2 Specific Objectives

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2 THEORETICAL BASIS

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3 CONCLUSION

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REFERENCES

ACEITUNO, José María; GUASQUE, Ana; BALBASTRE, Patricia; BLANES, Francisco; POMANTE, Luigi. Optimized scheduling of periodic hard real-time multicore systems. **IEEE Access**, IEEE, v. 11, p. 30027–30039, 2023.

DAS, Sanjeev; WERNER, Jan; ANTONAKAKIS, Manos; POLYCHRONAKIS, Michalis; MONROSE, Fabian. Sok: The challenges, pitfalls, and perils of using hardware performance counters for security. *In:* IEEE. 2019 IEEE symposium on security and privacy (SP). [*S. l.: s. n.*], 2019. p. 20–38.

GRACIOLI, Giovani; TABISH, Rohan; MANCUSO, Renato; MIROSANLOU, Reza; PELLIZZONI, Rodolfo; CACCAMO, Marco. Designing mixed criticality applications on modern heterogeneous mpsoc platforms. *In:* SCHLOSS DAGSTUHL–LEIBNIZ-ZENTRUM FUER INFORMATIK. 31ST Euromicro Conference on Real-Time Systems (ECRTS 2019). [*S. l.: s. n.*], 2019. p. 27–1.

HOFFMANN, José Luis Conradi; FRÖHLICH, Antônio Augusto. Online machine learning for energy-aware multicore real-time embedded systems. **IEEE Transactions on Computers**, IEEE, v. 71, n. 2, p. 493–505, 2021.

HORSTMANN, Leonardo Passig; HOFFMANN, José Luis Conradi; FRÖHLICH, Antônio Augusto. A framework to design and implement real-time multicore schedulers using machine learning. *In:* IEEE. 2019 24th IEEE international conference on emerging technologies and factory automation (ETFA). [*S. l.: s. n.*], 2019. p. 251–258.

ISLAM, Fakhruddin Muhammad Mahbub ul; LIN, Man; YANG, Laurence T; CHOO, Kim-Kwang Raymond. Task aware hybrid DVFS for multi-core real-time systems using machine learning. **Information Sciences**, Elsevier, v. 433, p. 315–332, 2018.

JADON, Shruti; KANNAN, Pradyuman Kavedi; KALARIA, Urmil; VARSHA, KR; GUPTA, Karthik; HONNAVALLI, Prasad B. A comprehensive study of load balancing approaches in real-time multi-core systems for mixed real-time tasks. **IEEE Access**, IEEE, v. 12, p. 53373–53395, 2024.

LUGO, Tamara; LOZANO, Santiago; FERNÁNDEZ, Javier; CARRETERO, Jesus. A survey of techniques for reducing interference in real-time applications on multicore platforms. **IEEE Access**, IEEE, v. 10, p. 21853–21882, 2022.

MRABET, Faten; KARAMTI, Walid; MAHFOUDHI, Adel. Scheduling analysis and correction for dependent real-time tasks upon heterogeneous multiprocessor architectures. **Computing**, Springer, v. 106, n. 3, p. 651–712, 2024.

RAHIM, Shaima; NAIR, Piyoosh Purushothaman; DEVARAJ, Rajesh. A survey of machine learning-driven task scheduling approaches for multiprocessor systems. **Journal of Systems Architecture**, Elsevier, p. 103628, 2025.

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